

Lecture 7: Interpreting Regressions and Regression Tables

We have now covered the theory of estimation for ordinary least squares. Today we will cover a few more practical issues. First, we will discuss a few common variable transformations that are useful in regression, as well as how to interpret them. We will then turn to interpreting regression output from Stata, as well as regression tables of the type that you might find in an economics paper. In the process, we will also discuss the R^2 and the root mean squared error (RMSE) of a regression.

Variable Transformations

Our standard regression model fits the outcome, Y_i , as a linear function of one or more regressors, $X_{i,1}, \dots, X_{i,K}$. This linear framework can accommodate a number of non-linear models through variable transformations. We discuss a few of these, and their interpretation, below.

Taking Logs

We will use $\log(\cdot)$ to denote the natural logarithm. We can take logs of the outcome and/or of the regressor.

Log Outcome First, consider the case where we take the log of the outcome. We run the regression:

$$\log Y_i = \alpha + \beta \cdot X_i + \varepsilon_i$$

where X_i is our regressor of interest, and ε_i satisfies the usual orthogonality condition. Note that this model is equivalent to writing:

$$Y_i = e^{\alpha + \beta \cdot X_i + \varepsilon_i}$$

What does β reflect in our model? We have

$$\beta = \frac{\partial \log Y_i}{\partial X_i} = \frac{\partial Y_i}{\partial X_i} \cdot \frac{1}{Y_i}$$

which tells us that β reflects the proportional change in Y_i per unit change in X_i . In economics, this is often called a semi-elasticity.

Log Points vs. Percent For example, we might run a regression of log wages on years of schooling. Suppose we estimated $\beta = 0.08$. We often interpret that regression by saying that “an additional year of schooling raises wages by 8%.” Note, however, that this is an approximation, and is not quite right. Suppose that we look at someone with 12 years of schooling, and we want to compute how wages would change at other levels of schooling. Let $Y_i(12) = e^{\alpha + \beta \cdot 12 + \varepsilon_i}$ denote their wages at 12 years of schooling. At 13 years of schooling, they would earn:

$$e^{\alpha + \beta \cdot 13 + \varepsilon_i} = e^{\beta} \cdot Y_i(12) = e^{0.08} \cdot Y_i(12) \approx 1.0833 \cdot Y_i(12)$$

so their wages would actually rise by 8.33% rather than 8%. At 11 years of schooling, they would earn $e^{-0.08} \cdot Y_i(12) \approx 0.9231$, so their wages would go down by 7.69%. Thus, although our percent approximation is pretty good, it is not exact, and will break down for larger changes. If we want to be precise, we can instead use the concept of “log points,” and say “an additional year of schooling raises wages by 8 log points.” Log points are in some ways actually nicer than percent changes, because log points can be added and subtracted: an increase of 50 log points followed by a decrease of 50 log points ends up exactly where you started, but a 50% increase followed by a 50% decrease will end up at 75% of the original value.

Log Regressor We can also take the log of our regressor. The interpretation is similar: a 0.01 change in $\log X_i$ translates almost exactly to a 1% change in X_i , but the approximation between log changes and percent changes breaks down for larger changes.

If we take the log of both the regressor and the outcome, then we can interpret the coefficient as an elasticity. For example, in the following regression:

$$\log \text{Quantity}_i = \alpha + \beta \cdot \log \text{Price}_i + \varepsilon_i$$

the coefficient β could be interpreted as the price elasticity of demand (of course, whether this estimates demand or supply will depend on which orthogonality condition holds!).

Reasons to Take Logs The main reason to take logs is to give the regression coefficient a percent change interpretation. Relatedly/equivalently, we sometimes think that the appropriate functional form is multiplicative rather than additive: for example, if we think there is a Cobb-Douglas production function, $Y_i = z_i \cdot L_i^{\beta_L} K_i^{\beta_K}$, then we can take logs and estimate $\log Y_i = \alpha + \beta_L \log L_i + \beta_K \log K_i + \varepsilon_i$. Taking logs can also be useful if the variable's distribution has a long right tail: a regression with income as an outcome might be driven by a few high earners, leading to high variance. Taking logs will downweight the right tail, and may lead to more stable estimates.

Dummy Variables

Often, we are interested in categorical variables: variables that assign observations to groups, rather than numeric variables. To put these into a regression, we need to turn these categories into numbers.

We do this with dummy variables: indicators that are equal to one if the observation is in the group, and zero if it is not in the group. In general, if we have K mutually exclusive groups, then we include $K - 1$ dummy variables. We must omit one category to avoid the “dummy variable trap.” The dummy variable trap is the fact that if we had dummy variables for all K groups, then we would have perfect collinearity: one of the dummy variables would be equal to one minus the sum of the other dummy variables. Since we cannot have perfect collinearity in our regression, we must omit a category.

For example, suppose that we are studying people's flavor rating of three categories of ice cream: chocolate, vanilla, and strawberry. We would run the regression:

$$\text{Flavor Rating}_i = \alpha + \beta_{\text{Chocolate}} \cdot \text{Chocolate}_i + \beta_{\text{Vanilla}} \cdot \text{Vanilla}_i + \varepsilon_i$$

where Chocolate_i is an indicator equal to one if the flavor being tested is chocolate, Vanilla_i is an indicator equal to one if the flavor being tested is vanilla, and where the omitted category is strawberry. Since strawberry is the omitted category, α will be the average flavor rating of strawberry, $\alpha + \beta_{\text{Vanilla}}$ is the average flavor rating of vanilla, and $\alpha + \beta_{\text{Chocolate}}$ is the average flavor rating of chocolate.

We might also include dummy variables as controls: these variables will shift the intercept of the regression. Returning to our height and earnings example from Case and Paxson (2008), we might estimate the effect of height on earnings, controlling for gender:

$$\text{Earnings}_i = \alpha + \beta_{\text{Height}} \cdot \text{Height}_i + \beta_{\text{Female}} \cdot \text{Female}_i + \varepsilon_i$$

where Female_i is a dummy that is equal to one if the observation is female. Note that in a regression like this, it makes no difference whether we make the omitted category male or female: we will get the same estimate of β_{Height} .

Interactions

Sometimes we are interested not just in controlling for dummy variables, but in knowing how an effect varies across different groups or across individuals with different values of certain regressors. To do this, we can include an interaction effect. Continuing with the earnings example, we can run:

$$\text{Earnings}_i = \alpha + \beta_{\text{Height}} \cdot \text{Height}_i + \beta_{\text{Female}} \cdot \text{Female}_i + \gamma \cdot \text{Height}_i \cdot \text{Female}_i + \varepsilon_i$$

Here, since male is the excluded category, β_{Height} now reflects the slope from regressing earnings on height for men, while $\beta_{\text{Height}} + \gamma$ is the slope from regressing earnings on height for women. We say that β_{Height} and β_{Female} are the main effects, while γ is the interaction effect. Note that whenever we run a regression with an interaction effect, it is always essential to include the main effects. Otherwise, the interaction term would be an (uninterpretable) mixture of the missing main effect and of the interaction term.

Polynomials

One more way that we can incorporate nonlinearities into a linear regression is through polynomials. If we are interested in the nonlinear effect of X_i on Y_i , then we can include higher-order terms in our regression:

$$Y_i = \alpha + \beta_1 \cdot X_i + \beta_2 \cdot X_i^2 + \beta_3 \cdot X_i^3 + \varepsilon_i$$

The marginal effect of X_i on Y_i will then depend on the value of X_i . Here, we have $\frac{\partial Y_i}{\partial X_i} = \beta_1 + 2 \cdot \beta_2 \cdot X_i + 3 \cdot \beta_3 \cdot X_i^2$.

Reading Regression Output

We will now look at actual regression output from Stata. To generate these regressions, I used the WAGE1.dta file from Wooldridge; this is an extract from the 1976 Current Population Survey. We run a regression of log wages on years of education and a dummy variable for female. The regression is:

$$\log \text{ wage}_i = \alpha + \beta_1 \cdot \text{Years of Schooling}_i + \beta_2 \cdot \text{Female}_i + \varepsilon_i$$

The regression output is:

```
. reg log_wage educ female, rob
```

Linear regression	Number of obs	=	526
	F(2, 523)	=	105.65
	Prob > F	=	0.0000
	R-squared	=	0.3002
	Root MSE	=	.4455

		Robust				
log_wage	Coefficient	std. err.	t	P> t	[95% conf. interval]	
educ	.0772033	.0075908	10.17	0.000	.0622912	.0921155
female	-.3608654	.0387863	-9.30	0.000	-.4370615	-.2846694
_cons	.8262694	.1000852	8.26	0.000	.629651	1.022888

Let's go through everything in this regression output.

First, `reg log_wage educ female, rob` is a command that runs our regression. Note that we include the option `rob`, which is short for robust, and tells Stata to use robust standard errors. You should always tell Stata to use robust standard errors; we will talk about this more in a future lecture.

The main table shows the coefficients and standard errors. The row `educ` shows the coefficient on years of schooling, β_1 , the row `female` shows the coefficient on female, β_2 , and the row `_cons` shows the intercept, α . The first column shows coefficients, the next column shows standard errors, and the next column shows t-statistics: the ratio of the coefficient to the standard error. This t-statistic can be used to test the null hypothesis that the coefficient is zero, and the following column shows the p-value on that null hypothesis, using the t-test. Finally, the last two columns show 95% confidence intervals for each coefficient.

In the top right, we see the number of observations, followed by a test statistic that tests the null hypothesis that all of the coefficients other than the intercept are zero. The output shows the "F-statistic" as well as the associated p-value. We will discuss this further in a future lecture.

Table 1: Relationship Between Log Wages and Education

	(1)	(2)	(3)	(4)
	Baseline	+ Female	+ Educ × Female	Quadratic in Educ
Education (Years)	0.083*** (0.008)	0.077*** (0.008)	0.077*** (0.009)	-0.067*** (0.025)
Female		-0.361*** (0.039)	-0.360* (0.203)	
Education × Female			-0.000 (0.016)	
Education ²				0.006*** (0.001)
Constant	0.584*** (0.098)	0.826*** (0.100)	0.826*** (0.117)	1.408*** (0.132)
R ²	0.186	0.300	0.300	0.216
Observations	526	526	526	526

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Finally, the last two rows in the top right show the R^2 and the root mean squared error (RMSE). These are two closely related measures of the “fit” of the regression. The R^2 measures how much of the variance of Y_i is explained by the regressors:

$$R^2 = \frac{\text{Var}(\hat{Y}_i)}{\text{Var}(Y_i)} = 1 - \frac{\text{Var}(\hat{\varepsilon}_i)}{\text{Var}(Y_i)}$$

The RMSE is the standard deviation of the estimated residual $RMSE = \sqrt{\text{Var}(\hat{\varepsilon}_i)}$.¹

In this example, the R^2 is 30%, which is quite high for social sciences. But it is important to note that having a high or low R^2 says little about whether the regression is good or estimates a causal effect. Often, effects are small relative to the wide variety of factors that affect the outcome, leading to a small R^2 . But what matters for causality is whether the error term is orthogonal to the regressor, not the R^2 .

Reading Regression Tables

Now, let’s use the same data set to generate a regression table. The results are in Table 1.

This is a fairly standard looking regression table, which we can use to learn how to read a typical regression table. Each column represents a different regression. You can infer what regressors were included in each specification based on which rows are populated for that column. The “baseline” specification is the first one, and later columns add additional controls or interaction effects. The coefficient estimate is shown without parentheses, and the standard errors are in parentheses below the coefficient estimate. Additional statistics are reported at the bottom of the table (here, those are the R^2 and the number of observations, but in other contexts there could be other things). Note also that there are stars on certain estimates, reflecting that those estimates are statistically significantly different from zero: the number of stars tells us the level of significance. This practice of adding stars to tables is getting less common, as many economists think that it encourages focusing on arbitrary thresholds for statistical significance, rather than on more important questions. However, you will still see it, especially on older papers, so I have done it in this example.

¹Stata uses a “degrees-of-freedom” correction when reporting the RMSE, multiplying the sample standard deviation by $\sqrt{\frac{N}{N-K-1}}$, where K is the number of regressors (other than the intercept).